THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon, and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** saprotrophs	e.g. mushrooms
2	**(c)** all day and night	all living cells
3	**(d)** the stalk joining leaf to stem	part of leaf
4	**(c)** parasitic	host harmed
5	**(b)** vacuum (space)	light needs no medium
6	**(d)** d-like (laterally inverted)	left-right swap
7	**(d)** lateral inversion	e.g. AMBULANCE
8	**(c)** reflection	law of reflection
9	**(d)** 23.5°	from vertical to orbit plane
10	**(a)** eyes	retina
11	**(b)** equinoxes	21 March and 23 September
12	**(d)** 24 hours	one day

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Remove chlorophyll in warm alcohol (1); add iodine, blue-black shows starch (1).	1+1
14	Light source, opaque object, screen (2).	2
15	Light travels in straight lines and is blocked (2).	2
16	Motion around the Sun (1); one year (1).	1+1
17	Days when day and night are equal (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Pitcher-shaped leaf with slippery rim (1.5); insects fall and are digested (1.5).	1½+1½
19	Many rays from one point reach different places (1.5); images overlap (1.5).	1½+1½
20	Tilt changes direction of sunlight (1.5); more direct rays give summer (1.5).	1½+1½
21	Use certified filters (1.5); pinhole projection (1.5).	1½+1½
22	Leaves make food (1); phloem carries to all parts (1); stored in roots, fruits (1).	1+1+1
	OR: Cover part of a leaf with black paper (1); expose to sun, remove chlorophyll and add iodine (1); covered part shows no starch (1).	1+1+1
23	Shadow: dark patch from blocked light (1.5); image: inverted copy of the object's shape and colours (1.5).	1½+1½
	OR: Image is 30 cm behind the mirror (1.5); total distance 60 cm (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Ball blocks light (1). (ii) Shadow grows larger (1). (iii) Lighter shadow (1). (iv) Straight-line travel (2).	1+1+1+2
	OR: (i) Own light vs reflected (2). (ii) Sun, bulb; Moon, book (2). (iii) Reflects sunlight (1).	2+2+1
25	(i) Spin; orbit (2). (ii) 24 hours; 365¼ days (2). (iii) Day-night; seasons (1).	2+2+1
	OR: (i) Relative positions (1). (ii) Waxing (1). (iii) Purnima (1). (iv) Amavasya (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon, and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** green plants	make own food
2	**(b)** starch	test for starch
3	**(c)** pale and weak	no photosynthesis
4	**(b)** guard cells	control opening
5	**(d)** an angle to each other	creates symmetry
6	**(c)** luminous	e.g. the Sun
7	**(a)** translucent	partly allows light
8	**(d)** opaque	blocks sunlight
9	**(b)** rotation	causes day and night
10	**(a)** from anywhere on the night side of Earth	safe to view
11	**(b)** natural satellite	of the Earth
12	**(b)** 365¼ days	one year

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Xylem carries water and minerals (1); phloem carries food (1).	1+1
14	Carbon dioxide + Water → Glucose + Oxygen, in sunlight and chlorophyll (2).	2
15	Virtual and erect (1); same size, equal distance behind mirror (1).	1+1
16	Earth's axis is tilted (1); and it revolves around the Sun (1).	1+1
17	Days with longest or shortest daylight (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Only green patches have chlorophyll (1.5); only they turn blue-black with iodine (1.5).	1½+1½
19	Xylem: water, minerals, upward (1.5); phloem: food, both directions (1.5).	1½+1½
20	Mirror is smooth and gives regular reflection (1.5); wall is rough and diffuses light (1.5).	1½+1½
21	Year divisible by 4 (1.5); 2024, 2028 (1.5).	1½+1½
22	Closer to the source makes a larger shadow (1.5); farther makes a smaller shadow (1.5).	1½+1½
	OR: Use three cards with holes in a line (1); light seen only when holes aligned (1); bend one card and light stops (1).	1+1+1
23	Earth between Sun and Moon (1); on full moon day (1); Earth's shadow on Moon (1).	1+1+1
	OR: New moon, waxing crescent (1); first quarter, full moon (1); waning phases back to new moon (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Role of light (1). (ii) To see iodine colour (1). (iii) Covered part no blue-black (1). (iv) Light is necessary (2).	1+1+1+2
	OR: (i) Water loss (1). (ii) Heat, wind, humidity (2). (iii) Cooling, pull (2).	1+2+2
25	(i) New moon (1). (ii) Sun, Moon, Earth (1). (iii) Filters/projection (1). (iv) Eye damage (2).	1+1+1+2
	OR: (i) Tilted orbit (2). (ii) A few times a year (1). (iii) Study of Sun's corona (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon, and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** nitrogen	poor soil
2	**(a)** phloem	translocation
3	**(a)** darker green and receives more light	palisade layer
4	**(a)** oxygen	by-product
5	**(c)** transparent	e.g. clear glass
6	**(d)** diffused reflection	scattering
7	**(d)** reflection	mirror
8	**(d)** blurred	overlap of rays
9	**(d)** Earth's rotation	daily motion
10	**(a)** near the extension of Earth's axis	Dhruva Tara
11	**(b)** 29.5 days	synodic month
12	**(a)** reflects sunlight	non-luminous

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Pulls water upward and cools the plant (2).	2
14	Tiny pores on leaves (1); exchange gases and release water vapour (1).	1+1
15	Angle of incidence equals angle of reflection (2).	2
16	Luminous give own light, e.g. Sun (1); non-luminous only reflect light, e.g. Moon (1).	1+1
17	Spinning on its axis (1); day and night (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Rhizobium fixes nitrogen (1.5); enriches soil (1.5).	1½+1½
19	Light travels straight (1); opaque object blocks it (1); dark region on screen (1).	1+1+1
20	Virtual, erect (1); same size (1); laterally inverted, equal distance behind (1).	1+1+1
21	Solar: Moon blocks Sun, new moon (1.5); lunar: Earth's shadow on Moon, full moon (1.5).	1½+1½
22	Moon between Sun and Earth (1); on new moon day (1); shadow falls on Earth (1).	1+1+1
	OR: Earth revolves (1.5); we face different parts of space (1.5).	1½+1½
23	Makes food for all (1.5); releases oxygen (1.5).	1½+1½
	OR: Parasitic (1); insectivorous (1); saprotrophic (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Making food using sunlight (1). (ii) CO_2, water → glucose, O_2 (2). (iii) Chloroplasts in leaves (2).	1+2+2
	OR: (i) Cuscuta (1). (ii) Parasitic (1). (iii) Weakens host (1). (iv) Insectivorous or saprotrophic (2).	1+1+1+2
25	(i) Left-right reversed (1). (ii) Lateral inversion (1). (iii) 40 cm behind the mirror (1). (iv) Virtual, erect, same size (2).	1+1+1+2
	OR: (i) Box, hole, screen (2). (ii) Inverted (1). (iii) Larger hole gives blurred, brighter image (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.